# How AI Chatbot Response Style Affects Cognitive Load and Performance in Educational Tasks

Research Proposal

**Selected Paper's Abstract.** Srinivasaprasad et al. (2026) showed that AI chatbot response style significantly affects students' cognitive load and task accuracy, and identified concise, structured responses as the optimal style for reducing mental effort. However, their study used a static, scripted (Wizard-of-Oz) chatbot tested in a single session, leaving open whether the same low-effort response style helps or quietly harms students' independent thinking over repeated use. This proposal outlines a research direction that builds a real, adaptive chatbot on the same experimental foundation and tests whether minimizing cognitive load can be achieved *without* inducing over-reliance or weakening critical thinking.

## 1. Background

Generative AI chatbots are now widely used in educational settings to support learning. A growing body of Human-Computer Interaction (HCI) research shows that these tools can reduce the mental effort (**cognitive load**) students experience while solving problems. At the same time, when AI makes tasks feel too easy, students may begin to **over-rely** on it, gradually weakening their own **critical thinking** and reasoning skills. This creates a natural tension: the same design choices that make an AI system easier to use may be the ones that make students think less for themselves.

This proposal is anchored in one foundational paper that directly measured this tension in a controlled setting.

### Anchor Paper

**Srinivasaprasad, A., Sepanloo, K., Naderian Jahromi, S., Yu, D., & Duffy, V. G. (2026).** *How AI Chatbot Response Style Affects Cognitive Load and Performance in Educational Tasks.* HCII 2025, LNCS 16339, pp. 188–200.

## 2. What the Anchor Paper Did and Found

- **Design:** 31 university students solved GRE-style verbal reasoning questions, first without AI help (baseline), then with AI help delivered through 4 scripted response styles: *Standard* (concise, structured), *Long* (detailed paragraphs), *Unstructured* (bottom-up list), and *Ambiguous* (vague, indirect).

- **Measures:** Physiological cognitive load via eye-tracking (pupil diameter, fixation duration) combined with subjective post-task workload surveys.
- **Key results:**
  - AI assistance raised accuracy from **30.6% to 74.0%**.
  - **Standard** responses produced the highest accuracy and the lowest reported mental effort and frustration.
  - **Unstructured** and **Ambiguous** responses produced the lowest accuracy and highest cognitive strain.
  - Self-reported mental effort was inversely correlated with accuracy ($\rho = -0.52$, $p = .0025$).
- **Conclusion of the paper:** Chatbot interfaces should default to concise, structured, low-load response styles to support learning.

## 3. Gaps Identified in the Paper (Self-Acknowledged Limitations)

The authors explicitly flag three limitations that leave the central design question unresolved:

1. **Static, scripted responses (Wizard-of-Oz design).** The chatbot did not actually generate responses; researchers manually selected pre-written answers. This cannot capture how a real, adaptive AI system would behave or how users would respond to it.
2. **No adaptive system was built or tested.** The authors propose, as future work, a *dynamic* AI that adjusts its response style in real time based on the user's physiological state but they never design or test such a system themselves.
3. **Single-session, short-term measurement only.** The study captured immediate cognitive load and immediate accuracy in one sitting. It does not examine retention, learning transfer, or most importantly for this proposal **changes in trust, reliance, or independent reasoning over repeated use**.

### The Deeper, Unstated Gap

The paper treats the *Standard* (low-load, high-accuracy) response style as the "winning" design. But a low-effort response is, by definition, the style that asks the least of the student's own thinking. The paper never asks whether the very style it recommends could, with repeated use, be the one most likely to encourage passive acceptance of AI answers i.e., **over-reliance** and a gradual decline in **critical thinking**. This is a well-documented risk in the broader HCI literature on AI reliance and cognitive offloading, but it was never tested here because the study only measured a single session with a non-adaptive chatbot.

## 4. Research Question

*Does an AI chatbot response style that minimizes cognitive load and maximizes short-term accuracy (the "Standard" style identified as optimal by Srinivasaprasad et al.) lead to increased over-reliance and reduced critical thinking with repeated use and can an adaptive chatbot be designed that keeps cognitive load low across sessions* without *sacrificing students' independent reasoning?*

### Sub-Questions

1. **RQ1:** Does a real, adaptive AI system (replacing the original Wizard-of-Oz script) preserve the cognitive-load benefits found for Standard-style responses?
2. **RQ2:** Does repeated exposure to low-load, high-accuracy AI responses increase behavioral indicators of over-reliance (e.g., reduced independent attempt-rate, faster uncritical acceptance of AI answers)?
3. **RQ3:** Can an adaptive response design (e.g., one that occasionally prompts the student to reason first, or adjusts structure based on engagement signals) maintain low cognitive load *while* sustaining or improving critical thinking performance over multiple sessions?

## 5. How the Research Question Solves the Gap

| Gap in Anchor Paper | How Our RQ Addresses It | Resulting Contribution |
|---|---|---|
| Static, scripted (Wizard-of-Oz) responses | Build and test a real adaptive AI chatbot | Tests findings under real generative-AI conditions |
| No adaptive/dynamic system tested | Design an adaptive response mechanism as proposed (but never built) by the authors | Directly fulfills the authors' own suggested future work |
| Single-session, short-term measurement | Extend the design to repeated sessions | Reveals whether "optimal" low-load responses cause harm over time |
| No reliance / critical-thinking measures | Add behavioral and performance measures of over-reliance and independent reasoning | Connects cognitive load research to the trust/critical-thinking literature |

## 6. Significance

This research extends a well-designed but limited experimental study into a longitudinal, ecologically valid investigation of a trade-off that current HCI literature has identified but not resolved: **AI systems that are easiest on the mind may be the ones most likely to erode independent thinking over time.** By directly building the adaptive system the anchor paper's authors proposed but did not create, this work both validates and extends their findings, while contributing actionable design guidance for educational AI systems that support learning without undermining it.

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**Anchor reference:** Srinivasaprasad, A., Sepanloo, K., Naderian Jahromi, S., Yu, D., & Duffy, V. G. (2026). How AI Chatbot Response Style Affects Cognitive Load and Performance in Educational Tasks. In V. G. Duffy (Ed.), *HCII 2025*, LNCS 16339, pp. 188–200. Springer. https://doi.org/10.1007/978-3-032-13012-9_14